

CEMRACS,
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Approximation and learning with tensor networks

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Tensor networks are prominent tools for the representation or approximation of multivariate functions or multidimensional arrays.

- A long history in quantum physics.
- **Tree tensor networks** appeared independently in numerical analysis, as an extension of low-rank decompositions to high-order tensors.
- Growing use in statistics, data science and probabilistic modelling.

- For the **approximation of a known tensor** u with respect to a certain norm, we aim at finding a tensor network v with low complexity that minimizes

$$\|u - v\|.$$

Here, the aim is the **compression** of u or the **extraction of information** from u (data analysis).

- For the **solution of an equation** $Au = b$ (e.g. in quantum physics, uncertainty quantification, stochastic calculus), we aim at finding a tensor network v with low complexity that minimizes some distance to u , e.g. some residual norm

$$\|Av - b\|.$$

The aim is here to obtain an **approximation of the solution** u with a **low computational complexity**.

Computing with tensor networks

- In **tensor completion**, knowing some entries $(u(i))_{i \in \Omega}$ of a multidimensional array, we try to find a tensor network that suitably fit the data, e.g. by minimizing

$$\sum_{i \in \Omega} |u(i) - v(i)|^2,$$

The aim is here to **recover (or complete) a tensor from partial information**, by exploiting low-rank structures of the tensor.

- For **inverse problems**, we want to identify a tensor u from indirect and partial observations $y = Au$ or $y = Au + \epsilon$, where A is an observation map. We try to find a tensor network that suitably fit the observations by minimizing some distance

$$d(y, Av)$$

between observations and the prediction Av .

Exploiting low-rank structures in u allows to reduce the number of parameters to estimate and possibly **makes the problem well-posed**.

- Approximating a function u from evaluations $u(x^k)$ at some points x^k , e.g. by minimizing

$$\frac{1}{n} \sum_{k=1}^n (u(x^k) - v(x^k))^2.$$

Depending on the context, points can be given or chosen. Here we want to **exploit at best the given evaluations** or **obtain a good approximation using a small number of evaluations**.

Computing with tensor networks

- In supervised or unsupervised learning, tensor networks are used as a **powerful model class** for high-dimensional tasks.
- **Supervised learning** of the relation between a random variable Y and another random variable X . Introduction of a risk functional

$$\mathcal{R}(v) = \mathbb{E}(\ell(Y, v(X)))$$

that quantifies some expected distance between observations Y and predictions $v(X)$. In practice, using samples $\{(x_k, y_k)\}_{k=1}^n$, we optimize an empirical risk

$$\frac{1}{n} \sum_{k=1}^n \ell(y^k, v(x^k))$$

- **Estimation of the density** of a random variable X from samples $\{x_k\}_{k=1}^n$. If the density u minimizes some functional

$$\mathcal{R}(v) = \mathbb{E}(\gamma(v, X)),$$

we minimize in practice an empirical risk

$$\frac{1}{n} \sum_{k=1}^n \gamma(v, x^k)$$

Outline of the course

- Part I: Tensors, ranks and tensor networks
- Part II: Approximation theory of tree tensor networks
- Part III: Computational aspects

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Approximation and learning with tensor networks

Part I: Tensors, ranks and tensor networks

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Outline

- 1 Tensors
- 2 Tensor ranks
- 3 Tensor networks
- 4 Tensorization

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Algebraic tensors

Given d index sets $I_\nu = \{1, \dots, N_\nu\}$, $1 \leq \nu \leq d$, we introduce the multi-index set

$$I = I_1 \times \dots \times I_d.$$

An element v of the vector space $\mathbb{R}^I$ is a **tensor of order d** .

Algebraic tensors

Given d index sets $I_\nu = \{1, \dots, N_\nu\}$, $1 \leq \nu \leq d$, we introduce the multi-index set

$$I = I_1 \times \dots \times I_d.$$

An element v of the vector space $\mathbb{R}^I$ is a **tensor of order d** .

It can be represented by a **multidimensional array**

$$(v_i)_{i \in I} = (v_{i_1, \dots, i_d})_{i_1 \in I_1, \dots, i_d \in I_d}$$

that contains the coefficients of v in the canonical basis of $\mathbb{R}^I$, also denoted

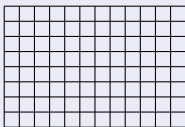
$$v(i) = v(i_1, \dots, i_d).$$

The order d is the number of **dimensions**, also known as **ways** or **modes**.

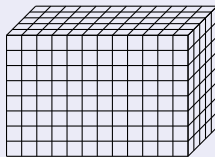
$d = 1$



$d = 2$



$d = 3$



Tensor diagram notations

A tensor is represented by a solid shape and tensor indices are notated by lines emanating from this shape.

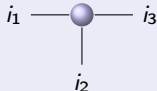
$d = 1$



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Connecting two index lines means contraction (or summation) over the corresponding indices.

$$i - \boxed{A} -^j \boxed{v} = \sum_j A(i, j) v(j)$$

Algebraic tensors

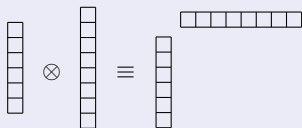
Given d vectors $v^{(\nu)} \in \mathbb{R}^{l_\nu}$, $1 \leq \nu \leq d$, the tensor product of these vectors

$$v := v^{(1)} \otimes \dots \otimes v^{(d)}$$

is called an **elementary tensor** and is such that

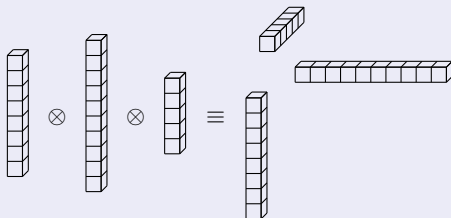
$$v(i) = v^{(1)}(i_1) \dots v^{(d)}(i_d)$$

$d = 2$



Using matrix notations, $v \otimes w$ is identified with the matrix vw^T .

$d = 3$



Algebraic tensors

The **tensor space** $\mathbb{R}^I = \mathbb{R}^{I_1 \times \dots \times I_d}$, also denoted $\mathbb{R}^{I_1} \otimes \dots \otimes \mathbb{R}^{I_d}$, is defined by

$$\mathbb{R}^I = \mathbb{R}^{I_1} \otimes \dots \otimes \mathbb{R}^{I_d} = \text{span}\{v^{(1)} \otimes \dots \otimes v^{(d)} : v^{(\nu)} \in \mathbb{R}^{I_\nu}, 1 \leq \nu \leq d\}$$

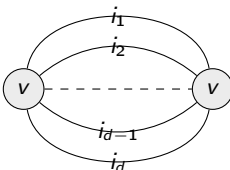
The **canonical norm on $\mathbb{R}^I$** , also called the **Frobenius norm**, is given by

$$\|v\| = \sqrt{\sum_{i \in I} v(i)^2}$$

and makes $\mathbb{R}^I$ a Hilbert space. It coincides with the **natural norm on $\ell_2(I)$** . It is the only norm associated with an inner product and having the crossnorm property

$$\|v^{(1)} \otimes \dots \otimes v^{(d)}\| = \|v^{(1)}\|_2 \dots \|v^{(d)}\|_2.$$

In tensor diagram notations

$$\|v\|^2 = \sum_{i \in I} v(i)^2 =$$


The diagram illustrates the Frobenius norm squared as a contraction of a tensor with itself. Two circular nodes, each labeled with the variable v , are connected by d arcs. The arcs are labeled with indices $i_1, i_2, \dots, i_d$ on the top half and i_{d-1}, i_d on the bottom half. A dashed horizontal line connects the two nodes, representing the summation over all indices in the set I .

Tensor product of functions

Let $V_\nu \subset \mathbb{R}^{\mathcal{X}_\nu}$ be a space of functions defined on $\mathcal{X}_\nu$.

$\mathcal{X}_\nu$ can be (a subset of) $\mathbb{R}$, $\mathbb{C}$, $\mathbb{N}$, $\mathbb{Z}$, or a set of vectors, sequences, graphs, images...

The tensor product of functions $v^{(\nu)} \in V_\nu$, denoted

$$v = v^{(1)} \otimes \dots \otimes v^{(d)},$$

is a multivariate function defined on $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_d$ and such that

$$v(x_1, \dots, x_d) = v^{(1)}(x_1) \dots v^{(d)}(x_d)$$

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Example

For $i \in \mathbb{N}_0^d$, the monomial $x^i = x_1^{i_1} \dots x_d^{i_d}$ is an elementary tensor.

Tensor product of functions

The **algebraic tensor product** of spaces V_ν is defined as

$$V_1 \otimes \dots \otimes V_d = \text{span}\{v^{(1)} \otimes \dots \otimes v^{(d)} : v^{(\nu)} \in V_\nu, 1 \leq \nu \leq d\}$$

which is the space of multivariate functions v which can be written as a finite linear combination of elementary (separated functions), i.e.

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A polynomial $\sum_i a_i x^i$ with $x^i = x_1^{i_1} \dots x_d^{i_d}$.

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A polynomial $\sum_i a_i x^i$ with $x^i = x_1^{i_1} \dots x_d^{i_d}$.

Up to a formal definition of the tensor product $\otimes$, the above construction can be extended to more general vector spaces (not only spaces of functions), including spaces of matrices or operators.

Infinite dimensional tensor spaces

For infinite dimensional spaces V_ν , a Hilbert (or Banach) tensor space equipped with a norm $\|\cdot\|$ is obtained by the completion (w.r.t. $\|\cdot\|$) of the algebraic tensor space

$$\overline{V^{\|\cdot\|}} = \overline{V_1 \otimes \dots \otimes V_d}^{\|\cdot\|}.$$

If the V_ν are Hilbert spaces with inner products $(\cdot, \cdot)_\nu$ and associated norms $\|\cdot\|_\nu$, a canonical inner product on V can be first defined for elementary tensors

$$(v^{(1)} \otimes \dots \otimes v^{(d)}, w^{(1)} \otimes \dots \otimes w^{(d)}) = (v^{(1)}, w^{(1)}) \dots (v^{(d)}, w^{(d)})$$

and then extended by linearity to the whole space V .

The associated norm $\|\cdot\|$ is called the **canonical norm**.

Example (L^p spaces)

Let $1 \leq p < \infty$. If $V_\nu = L_{\mu_\nu}^p(\mathcal{X}_\nu)$, then

$$L_{\mu_1}^p(\mathcal{X}_1) \otimes \dots \otimes L_{\mu_d}^p(\mathcal{X}_d) \subset L_\mu^p(\mathcal{X}_1 \times \dots \times \mathcal{X}_d)$$

with $\mu = \mu_1 \otimes \dots \otimes \mu_d$, and

$$\overline{L_{\mu_1}^p(\mathcal{X}_1) \otimes \dots \otimes L_{\mu_d}^p(\mathcal{X}_d)}^{\|\cdot\|} = L_\mu^p(\mathcal{X}_1 \times \dots \times \mathcal{X}_d)$$

where $\|\cdot\|$ is the natural norm on $L_\mu^p(\mathcal{X}_1 \times \dots \times \mathcal{X}_d)$.

Infinite dimensional tensor spaces

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where $\|\cdot\|$ is the natural norm on $L_\mu^p(\mathcal{X}_1 \times \dots \times \mathcal{X}_d)$.

Example (Bochner spaces)

Let $\mathcal{X}$ be equipped with a finite measure μ , and let W be a Hilbert (or Banach) space. For $1 \leq p < \infty$, the Bochner space $L_\mu^p(\mathcal{X}; W)$ is the set of Bochner-measurable functions $u : \mathcal{X} \rightarrow W$ with bounded norm $\|u\|_p = (\int_{\mathcal{X}} \|u(x)\|_W^p \mu(dx))^{1/p}$, and

$$L_\mu^p(\mathcal{X}; W) = \overline{W \otimes L_\mu^p(\mathcal{X})}^{\|\cdot\|_p}.$$

Infinite dimensional tensor spaces

Example (Sobolev spaces)

The Sobolev space $H^k(\mathcal{X})$ of functions defined on $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_d$, equipped with the norm

$$\|u\|_{H^k}^2 = \sum_{|\alpha|_1 \leq k} \|D^\alpha u\|_{L^2}^2,$$

is a Hilbert tensor space

$$H^k(\mathcal{X}) = \overline{H^k(\mathcal{X}_1) \otimes \dots \otimes H^k(\mathcal{X}_d)}^{\|\cdot\|_{H^k}}.$$

The Sobolev space $H_{mix}^k(\mathcal{X})$ equipped with the norm

$$\|u\|_{H_{mix}^k}^2 = \sum_{|\alpha|_\infty \leq k} \|D^\alpha u\|_{L^2}^2,$$

is a different tensor Hilbert space

$$H_{mix}^k(\mathcal{X}) = \overline{H^k(\mathcal{X}_1) \otimes \dots \otimes H^k(\mathcal{X}_d)}^{\|\cdot\|_{H_{mix}^k}}.$$

$\|u\|_{H_{mix}^k}$ is the canonical tensor norm on $H^k(\mathcal{X}_1) \otimes \dots \otimes H^k(\mathcal{X}_d)$.

Tensor product basis

If $\{\phi_i^{(\nu)}\}_{i \in I_\nu}$ is a basis of V_ν , then a basis of $V = V_1 \otimes \dots \otimes V_d$ is given by

$$\left\{ \phi_i = \phi_{i_1}^{(1)} \otimes \dots \otimes \phi_{i_d}^{(d)} : i \in I = I_1 \times \dots \times I_d \right\}.$$

A tensor $v \in V$ admits a decomposition

$$v = \sum_{i \in I} a_i \phi_i = \sum_{i_1 \in I_1} \dots \sum_{i_d \in I_d} a_{i_1, \dots, i_d} \phi_{i_1}^{(1)} \otimes \dots \otimes \phi_{i_d}^{(d)},$$

and v can be identified with the set of its coefficients

$$a \in \mathbb{R}^I.$$

Hilbert tensor spaces

If the $\{\phi_i^{(\nu)}\}_{i \in I_\nu}$ are orthonormal bases of spaces V_ν , then $\{\phi_i\}_{i \in I}$ is an orthonormal basis of the Hilbert tensor space $\overline{V}^{\|\cdot\|}$ equipped with the canonical norm. A tensor

$$v = \sum_{i \in I} a_i \phi_i$$

is such that

$$\|v\|^2 = \sum_{i \in I} a_i^2 := \|a\|^2.$$

Therefore, the map

$$a \mapsto \sum_{i \in I} a_i \phi_i$$

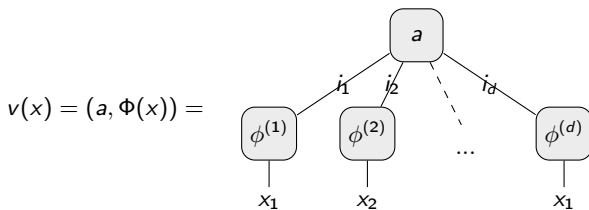
defines a linear isometry from $\ell_2(I)$ to V for finite dimensional spaces, and between $\ell_2(I)$ and $\overline{V}^{\|\cdot\|}$ for infinite dimensional spaces.

Tensor product feature map

If V is a space of functions defined on $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_d$, we introduce the feature map $\phi^{(\nu)}(x_\nu) = (\phi_{i_\nu}^{(\nu)}(x_\nu))_{i_\nu \in I_\nu} \in \mathbb{R}^{I_\nu}$ and the tensor product feature map $\Phi : \mathcal{X} \rightarrow \mathbb{R}^I$ such that

$$\Phi(x) = \phi^{(1)}(x_1) \otimes \dots \otimes \phi^{(d)}(x_d) \in \mathbb{R}^I$$

and a tensor v in V admits the representation



Outline

- 1 Tensors
- 2 Tensor ranks**
- 3 Tensor networks
- 4 Tensorization

Rank of order-two tensors

The **rank** of an order-two tensor $u \in V \otimes W$, denoted $\text{rank}(u)$, is the minimal integer r such that

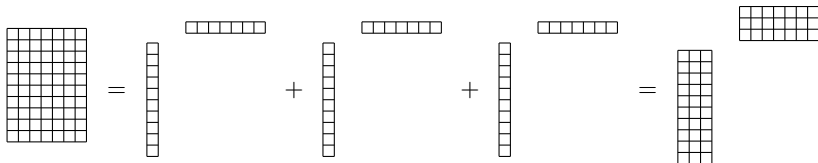
$$u = \sum_{k=1}^r v_k \otimes w_k$$

for some $v_k \in V$ and $w_k \in W$.

A tensor $u \in \mathbb{R}^n \otimes \mathbb{R}^m$ is identified with a matrix $u \in \mathbb{R}^{n \times m}$. The rank of u coincides with the **matrix rank**, which is the minimal integer r such that

$$u = \sum_{k=1}^r v_k w_k^T = VW^T,$$

where $V = (v_1, \dots, v_r) \in \mathbb{R}^{n \times r}$ and $W = (w_1, \dots, w_r) \in \mathbb{R}^{m \times r}$.



Singular value decomposition of order-two tensors

When V and W are Hilbert spaces (possibly infinite-dimensional), an algebraic tensor $u \in V \otimes W$ admits a singular value decomposition

$$u = \sum_{k \geq 1} \sigma_k v_k \otimes w_k,$$

where v_k and w_k are orthonormal vectors (singular vectors) and $\sigma_k \in \mathbb{R}^+$ are the singular values.

The rank of u is finite and coincides with the number of non-zero singular values,

$$\text{rank}(u) = \#\{k : \sigma_k \neq 0\}.$$

Example (Singular value decomposition of matrices)

For $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$, u is identified with a matrix in $\mathbb{R}^{n \times m}$ and

$$u = \sum_{k=1}^{\text{rank}(u)} \sigma_k v_k w_k^T = V S W^T$$

with orthogonal matrices V and W , and a diagonal matrix S .

Singular value decomposition of order-two tensors

An algebraic tensor $u \in V \otimes W$ can be identified with a linear operator from W to V with rank equal to $\text{rank}(u)$.

For infinite dimensional Hilbert spaces, the closure $\overline{V \otimes W}^{\|\cdot\|_V}$ of $V \otimes W$ with respect to the **injective norm** (corresponding to the **operator norm** or **spectral norm**) coincides with the space of compact operators.

A tensor $u \in \overline{V \otimes W}^{\|\cdot\|_V}$ still admits a **singular value decomposition**

$$u = \sum_{k \geq 1} \sigma_k v_k \otimes w_k.$$

and the rank (number of non-zero singular values) is possibly infinite.

Singular value decomposition of order-two tensors

Example (Proper Orthogonal Decomposition)

For $\Omega \times I$ a space-time domain and V a Hilbert space of functions defined on Ω , a function $u \in L^2(I; V)$ admits a singular value decomposition

$$u(t) = \sum_{k=1}^{\infty} \sigma_k v_k w_k(t)$$

which is known as the Proper Orthogonal Decomposition (POD).

Singular value decomposition of order-two tensors

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Example (Karhunen-Loeve decomposition)

For a probability space (Ω, μ) , an element $u \in L^2_{\mu}(\Omega; V)$ is a second-order V -valued random variable. If u is zero-mean, the singular value decomposition of u is known as the Karhunen-Loeve decomposition

$$u(\omega) = \sum_{k=1}^{\infty} \sigma_k v_k w_k(\omega)$$

where $w_k : \Omega \rightarrow \mathbb{R}$ are uncorrelated (orthogonal) random variables.

Low-rank format for order-two tensors

The set of tensors in $V \otimes W$ with rank bounded by r , denoted

$$\mathcal{R}_r = \{v : \text{rank}(v) \leq r\},$$

is **not a linear space nor a convex set**. However, it has **many favorable properties for a numerical use**.

- The application $v \mapsto \text{rank}(v)$ is lower semi-continuous, and therefore the set $\mathcal{R}_r$ is **closed**, which makes best approximation problems in $\mathcal{R}_r$ well posed.
- $\mathcal{R}_r$ is the **union of smooth manifolds** of tensors with fixed rank.

Canonical rank of higher-order tensors

For tensors $u \in V_1 \otimes \dots \otimes V_d$ with $d \geq 3$, there are different notions of rank.

The **canonical rank**, which is the natural extension of the notion of rank for order-two tensors, is the minimal integer r such that

$$u(x_1, \dots, x_d) = \sum_{k=1}^r v_k^{(1)}(x_1) \dots v_k^{(d)}(x_d),$$

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for some vectors $v_k^{(\nu)} \in V_\nu$.

Example

- A monomial $x^i = x_1^{i_1} \dots x_d^{i_d}$ has rank 1.
- A polynomial $\sum_{i \in \Lambda} a_i x^i$ has rank $\#\Lambda$.
- A Gaussian function $\exp(-\alpha \|x - a\|_2^2) = \prod_{i=1}^d \exp(-\alpha (x_i - a_i)^2)$ has rank 1.
- The function $\frac{1}{\|x\|_2}$ has infinite rank.

Canonical format

The subset of tensors in $V = V_1 \otimes \dots \otimes V_d$ with canonical rank bounded by r is denoted

$$\mathcal{R}_r = \{v \in V : \text{rank}(v) \leq r\}.$$

A tensor in $\mathcal{R}_r$ has a representation

$$v(x_1, \dots, x_d) = \sum_{k=1}^r v_k^{(1)}(x_1) \dots v_k^{(d)}(x_d)$$

The storage complexity of tensors in $\mathcal{R}_r$ is

$$\text{storage}(\mathcal{R}_r) = r \sum_{\nu=1}^d \dim(V_\nu) = O(rdn)$$

for $\dim(V_\nu) = O(n)$.

Canonical format

For $d \geq 3$, the set $\mathcal{R}_r$ loses many of the favorable properties of the case $d = 2$.

- Determining the rank of a given tensor is a NP-hard problem.

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Example

Consider the order-3 tensor

$$v = a \otimes a \otimes b + a \otimes b \otimes a + b \otimes a \otimes a$$

where a and b are linearly independent vectors in $\mathbb{R}^m$. The rank of v is 3. The sequence of rank-2 tensors

$$v_n = n\left(a + \frac{1}{n}b\right) \otimes \left(a + \frac{1}{n}b\right) \otimes \left(a + \frac{1}{n}b\right) - na \otimes a \otimes a$$

converges to v as $n \rightarrow \infty$.

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$$v_n = n\left(a + \frac{1}{n}b\right) \otimes \left(a + \frac{1}{n}b\right) \otimes \left(a + \frac{1}{n}b\right) - na \otimes a \otimes a$$

converges to v as $n \rightarrow \infty$.

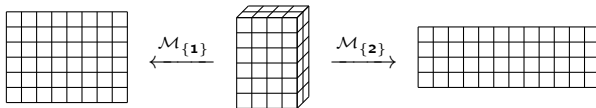
- The consequence is that for most problems involving approximation in canonical format $\mathcal{R}_r$, there is no robust method when $d > 2$.

α -rank

For a non-empty subset α of $D = \{1, \dots, d\}$, a tensor $u \in V = V_1 \otimes \dots \otimes V_d$ can be identified with an order-two tensor

$$\mathcal{M}_\alpha(u) \in V_\alpha \otimes V_{\alpha^c},$$

where $V_\alpha = \bigotimes_{\nu \in \alpha} V_\nu$, and $\alpha^c = D \setminus \alpha$. The operator $\mathcal{M}_\alpha : V \rightarrow V_\alpha \otimes V_{\alpha^c}$ is called the **matricisation (or unfolding) operator**.



The **α -rank** of u , denoted **$\text{rank}_\alpha(u)$** , is the rank of the order-two tensor $\mathcal{M}_\alpha(u)$,

$$\text{rank}_\alpha(u) = \text{rank}(\mathcal{M}_\alpha(u)),$$

which is the minimal integer r_α such that

$$\mathcal{M}_\alpha(u) = \sum_{k=1}^{r_\alpha} v_k^\alpha \otimes w_k^{\alpha^c}$$

for some $v_k^\alpha \in V_\alpha$ and $w_k^{\alpha^c} \in V_{\alpha^c}$. We note that **$\text{rank}_\alpha(u) = \text{rank}_{\alpha^c}(u)$** .

A multivariate function $u(x_1, \dots, x_d)$ with $\text{rank}_\alpha(u) \leq r_\alpha$ is such that

$$u(x) = \sum_{k=1}^{r_\alpha} v_k^\alpha(x_\alpha) w_k^{\alpha^c}(x_{\alpha^c})$$

for some functions $v_k^\alpha(x_\alpha)$ and $w_k^{\alpha^c}(x_{\alpha^c})$ of groups of variables

$$x_\alpha = \{x_\nu\}_{\nu \in \alpha} \quad \text{and} \quad x_{\alpha^c} = \{x_\nu\}_{\nu \in \alpha^c}.$$

Example

- $u(x) = u^1(x_1) \dots u^d(x_d)$ can be written $u(x) = u^\alpha(x_\alpha) u^{\alpha^c}(x_{\alpha^c})$, with $u^\alpha(x_\alpha) = \prod_{\nu \in \alpha} u^\nu(x_\nu)$. Therefore, for any α , $\text{rank}_\alpha(u) = 1$.

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- $u(x) = \sum_{k=1}^r u_k^1(x_1) \dots u_k^d(x_d)$ can be written $\sum_{k=1}^r u_k^\alpha(x_\alpha) u_k^{\alpha^c}(x_{\alpha^c})$ with $u_k^\alpha(x_\alpha) = \prod_{\nu \in \alpha} u_k^\nu(x_\nu)$. Therefore, for any α , $\text{rank}_\alpha(u) \leq r$, with equality if the functions $\{u_k^\alpha(x_\alpha)\}$ and the functions $\{u_k^{\alpha^c}(x_{\alpha^c})\}$ are linearly independent.

We deduce the following relation between α -ranks and canonical rank:

$$\text{rank}_\alpha(u) \leq \text{rank}(u), \quad \text{for any } \alpha.$$

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$$\text{rank}_\alpha(u) \leq \text{rank}(u), \quad \text{for any } \alpha.$$

- $u(x) = u^1(x_1) + \dots + u^d(x_d)$ can be written $u(x) = u^\alpha(x_\alpha) + u^{\alpha^c}(x_{\alpha^c})$, with $u^\alpha(x_\alpha) = \sum_{\nu \in \alpha} u^\nu(x_\nu)$. Therefore, $\text{rank}_\alpha(u) \leq 2$.

Example

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- $u(x) = \prod_{\alpha \in T} u^\alpha(x_\alpha)$ with T a collection of disjoint subsets, is such that $\text{rank}_\alpha(u) = 1$ for all $\alpha \in T$, and $\text{rank}_\gamma(u) \leq \prod_{\alpha \in T, \alpha \cap \gamma \neq \emptyset} \text{rank}_{\gamma \cap \alpha}(u^\alpha)$ for all γ .

α -ranks and minimal subspaces

For a subset α of $D = \{1, \dots, d\}$, the **minimal subspace**

$$U_{\alpha}^{min}(u)$$

of a tensor $u \in V_1 \otimes \dots \otimes V_d$ is defined as the **smallest subspace**

$$U_{\alpha} \subset V_{\alpha} = \bigotimes_{\nu \in \alpha} V_{\nu}$$

such that

$$\mathcal{M}_{\alpha}(u) \in U_{\alpha} \otimes V_{\alpha^c}.$$

The α -rank of u is the dimension of the minimal subspace $U_{\alpha}^{min}(u)$,

$$\text{rank}_{\alpha}(u) = \dim(U_{\alpha}^{min}(u)).$$

If u admits the representation

$$u(x) = \sum_{k=1}^{\text{rank}_{\alpha}(u)} v_k^{\alpha}(x_{\alpha}) v_k^{\alpha^c}(x_{\alpha^c})$$

then $U_{\alpha}^{min}(u) = \text{span}\{v_k^{\alpha} : 1 \leq k \leq \text{rank}_{\alpha}(u)\}$.

For any partition $\{\alpha_1, \dots, \alpha_m\}$ of D , an algebraic tensor u is such that

$$u \in U_{\alpha_1}^{\min}(u) \otimes \dots \otimes U_{\alpha_m}^{\min}(u)$$

Moreover, for any $\alpha \subset D$ and any partition $\{\beta_1, \dots, \beta_s\}$ of α , it holds

$$U_{\alpha}^{\min}(u) \subset U_{\beta_1}^{\min}(u) \otimes \dots \otimes U_{\beta_s}^{\min}(u)$$

that implies

$$\text{rank}_{\alpha}(u) \leq \prod_{k=1}^s \text{rank}_{\beta_k}(u)$$

Also, for any $p \in \{1, \dots, s\}$

$$\text{rank}_{\beta_p}(u) \leq \text{rank}_{\alpha}(u) \prod_{\substack{k=1 \\ k \neq p}}^s \text{rank}_{\beta_k}(u)$$

Example

The function

$$u(x_1, x_2, x_3) = \cos(x_1 + x_2) + x_1(x_2 + 2x_3) = \cos(x_1)\cos(x_2) - \sin(x_1)\sin(x_2) + x_1x_2 + 2x_1x_3$$

has for minimal subspaces and ranks

- $U_1^{\min}(u) = \text{span}\{\cos(x_1), \sin(x_1), x_1\}, \quad r_1 = 3$
- $U_2^{\min}(u) = \text{span}\{\cos(x_2), \sin(x_2), x_2, 1\}, \quad r_2 = 4$
- $U_3^{\min}(u) = \text{span}\{1, x_3\}, \quad r_3 = 2$
- $U_{1,2}^{\min}(u) = \text{span}\{\cos(x_1 + x_2) + x_1x_2, x_1\}, \quad r_{1,2} = 2$
- $U_{2,3}^{\min}(u) = \text{span}\{\cos(x_2), \sin(x_2), x_2 + 2x_3\}, \quad r_{2,3} = 3$
- $U_{1,3}^{\min}(u) = \text{span}\{\cos(x_1), \sin(x_1), x_1, x_1x_3\}, \quad r_{1,3} = 4$

In particular, we can check that

$$U_{1,3}^{\min}(u) \subset U_1^{\min}(u) \otimes U_3^{\min}(u) = \text{span}\{\cos(x_1), \sin(x_1), x_1, \cos(x_1)x_3, \sin(x_1)x_3, x_1x_3\}$$

$$r_{1,3} \leq r_1 r_3, \quad r_1 \leq r_{1,3} r_3, \quad r_3 \leq r_{1,3} r_1$$

Outline

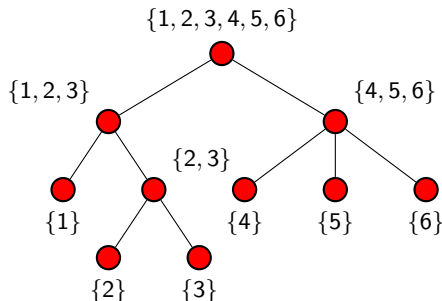
- 1 Tensors
- 2 Tensor ranks
- 3 Tensor networks**
- 4 Tensorization

Tree-based tensor format

Tree-based (Hierarchical) tensor formats [Hackbusch-Kuhn'09] are subsets of tensors

$$\mathcal{T}_r^T = \{v \in V : \text{rank}_\alpha(v) \leq r_\alpha, \alpha \in T\}$$

where $r = (r_\alpha)_{\alpha \in T}$ and where T is a **dimension partition tree** T over $D = \{1, \dots, d\}$, with root D and leaves $\mathcal{L}(T) = \{\{\nu\} : 1 \leq \nu \leq d\}$. All nodes in T are non empty subsets of D . The set of children of $\alpha \in T$ is either empty (for a leaf node) or is a nontrivial partition of α (for an interior node).



The **tree-based rank** of a tensor v is the tuple $\text{rank}_T(v) = (\text{rank}_\alpha(v))_{\alpha \in T}$.

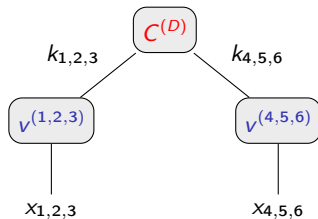
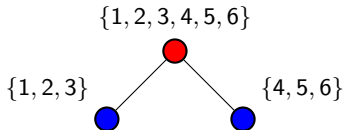
By convention, $\text{rank}_D(v) = 1$.

Tree-based tensor format

Elements of $\mathcal{T}_r^T$ admit an **explicit representation**. Let $v \in \mathcal{T}_r^T$ with T -rank $r = (r_\alpha)_{\alpha \in T}$. At the first level, v admits the representation

$$v(x) = \sum_{k_{\beta_1}=1}^{r_{\beta_1}} \dots \sum_{k_{\beta_s}=1}^{r_{\beta_s}} C_{k_{\beta_1}, \dots, k_{\beta_s}}^{(D)} v_{k_{\beta_1}}^{(\beta_1)}(x_{\beta_1}) \dots v_{k_{\beta_s}}^{(\beta_s)}(x_{\beta_s})$$

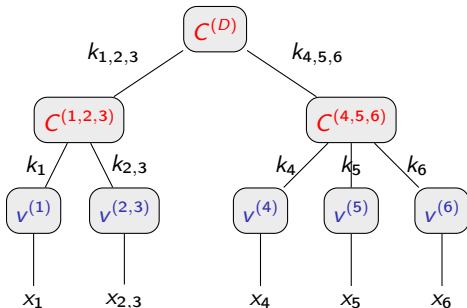
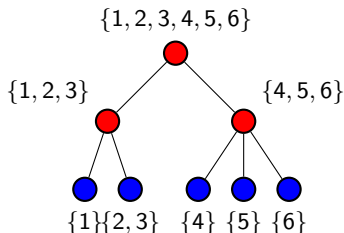
where $\{\beta_1, \dots, \beta_s\} = S(D)$ are the children of the root node D , and $\{v_{k_\beta}^{(\beta)}\}_{1 \leq k_\beta \leq r_\beta}$ form a basis of the minimal subspace $U_\beta^{\min}(v)$.



Tree-based tensor format

Then, for an interior node α of the tree, with children $S(\alpha) = \{\beta_1, \dots, \beta_s\}$, the functions (or tensors) $v_{k_\alpha}^{(\alpha)}$ admit the representation

$$v_{k_\alpha}^{(\alpha)}(x_\alpha) = \sum_{k_{\beta_1}=1}^{r_{\beta_1}} \dots \sum_{k_{\beta_s}=1}^{r_{\beta_s}} C_{k_\alpha, k_{\beta_1}, \dots, k_{\beta_s}}^{(\alpha)} v_{k_{\beta_1}}^{(\beta_1)}(x_{\beta_1}) \dots v_{k_{\beta_s}}^{(\beta_s)}(x_{\beta_s}).$$

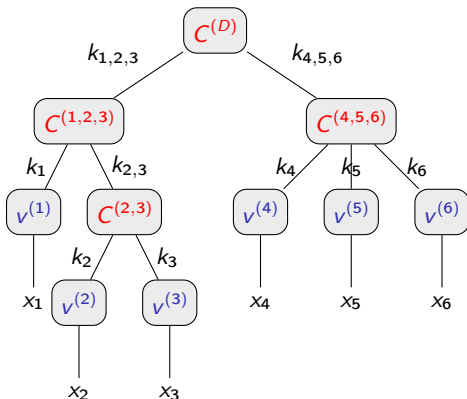
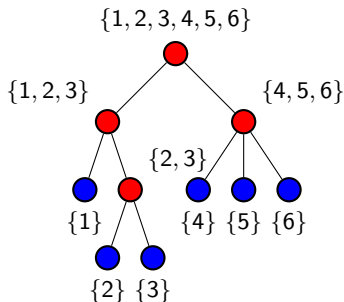


Tree-based tensor format as a tree tensor network

Finally, the tensor v admits the representation

$$v(x) = \sum_{\substack{1 \leq k_\beta \leq r_\beta \\ \beta \in T}} \prod_{\alpha \in T \setminus \mathcal{L}(T)} C_{(k_\beta)_{\beta \in S(\alpha)}, k_\alpha}^{(\alpha)} \prod_{\nu \in \mathcal{L}(T)} v_{k_\nu}^{(\nu)}(x_\nu)$$

where the parameters C^α and $v^{(\nu)}$ form a **tree tensor network**.

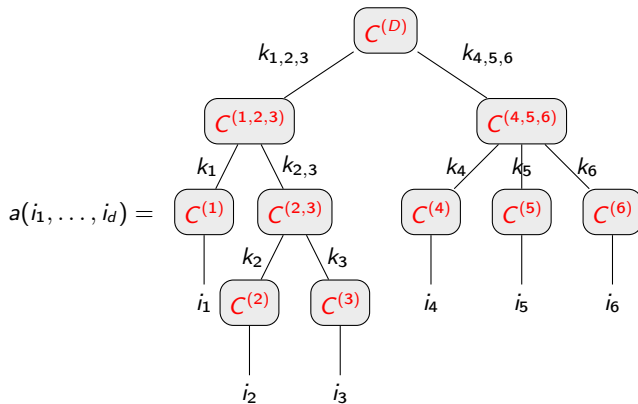


Tree-based tensor format as a tree tensor network

Given bases $\{\phi_{i_\alpha}^\alpha(x_\alpha)\}_{i_\alpha \in I^\alpha}$ of functions for the spaces V_α for $\alpha \in \mathcal{L}(T)$,

$$v(x) = \sum_{i_1 \in I^1} \dots \sum_{i_d \in I^d} a(i_1, \dots, i_d) \phi_{i_1}(x_1) \dots \phi_{i_d}(x_d)$$

with $a(i_1, \dots, i_d) = \sum_{\beta \in T} 1 \leq k_\beta \leq r_\beta \prod_{\alpha \in T \setminus \mathcal{L}(T)} C_{(k_\beta)_{\beta \in S(\alpha)}, k_\alpha}^{(\alpha)} \prod_{\alpha \in \mathcal{L}(T)} C_{i_\alpha, k_\alpha}^{(\alpha)}$ or using tensor diagram notations



Representation complexity

The representation complexity for the representation of a tensor in $\mathcal{T}_r^T(V)$ is

$$C(T, r) = \sum_{\alpha \in T \setminus \mathcal{L}(T)} r_\alpha \prod_{\beta \in S(\alpha)} r_\beta + \sum_{\nu \in \mathcal{L}(T)} \#I^\alpha r_\alpha.$$

If $r_\alpha = O(R)$ and $\#I^\alpha = O(N)$,

$$C(T, r) = O(dNR + (\#T - d - 1)R^{s+1} + R^s),$$

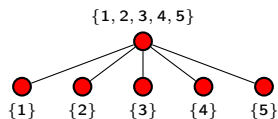
where $s = \max_{\alpha \in T \setminus \mathcal{L}(T)} \#S(\alpha)$ is the **arity** of the tree.

Since $\#T \leq 2d + 1$,

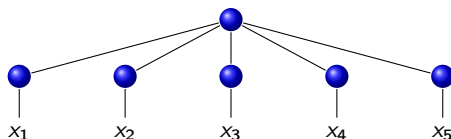
$$C(T, r) = O(dNR + dR^{s+1} + R^s)$$

Tucker format

The **Tucker format** [Hitchcock'27] corresponds to a **trivial tree** with one level, arity $s = d$, $\#T = d + 1$,



The representation of a tensor u in $\mathcal{T}_r^T$ is

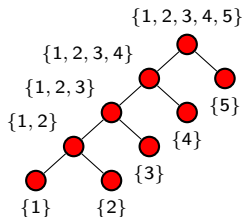


The representation complexity

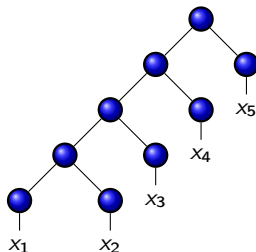
$$C(T, r) = O(dNR + R^d)$$

Tensor train Tucker format

The tensor train Tucker format corresponds to a **linear binary tree**



The representation of a tensor u in $\mathcal{T}_r^T$ is

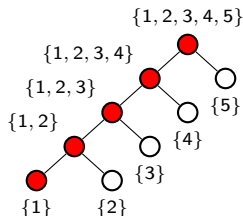


The representation complexity $C(T, r) = O(dNR + (d - 2)R^3 + R^2)$.

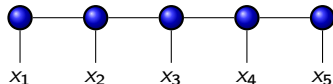
Tensor train format

The **tensor train format** [Oseledets-Tyrtysnikov'09] was discovered independently in quantum physics [Baxter'68 , Affleck'87] and coined **Matrix Product State** (MPS). It corresponds to a degenerate tree-based format where T is a subset of a linear tree

$$T = \{\{1\}, \{1, 2\}, \{1, 2, 3\}, \dots, \{1, \dots, d\}\}$$



The representation of a tensor u in $\mathcal{T}_r$ is



or explicitly

$$u(x_1, \dots, x_d) = \sum_{k_1=1}^{r_1} \dots \sum_{k_{d-1}=1}^{r_1, \dots, d-1} v_{k_1}^{(1)}(x_1) v_{k_1, k_2}^{(2)}(x_2) \dots v_{k_{d-2}, k_{d-1}}^{(d-1)}(x_{d-1}) v_{k_{d-1}}^{(d)}(x_d)$$

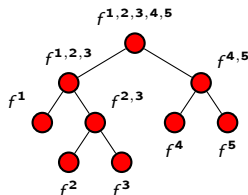
The complexity is $C(T, r) = O(dNR^2)$.

Tree tensor networks as a compositional function network

By identifying a tensor $\mathcal{C}^{(\alpha)} \in \mathbb{R}^{n_1 \times \dots \times n_s \times r_\alpha}$ with a $\mathbb{R}^{r_\alpha}$ -valued multilinear function

$$f^{(\alpha)} : \mathbb{R}^{n_1} \times \dots \times \mathbb{R}^{n_s} \rightarrow \mathbb{R}^{r_\alpha},$$

a function v in $\mathcal{T}_r^T$ admits a representation as a tree-structured composition of multilinear functions $\{f^{(\alpha)}\}_{\alpha \in T}$.



$$v(x) = f^D(f^{1,2,3}(f^1(\Phi^1(x_1)), f^{2,3}(f^2(\Phi^2(x_2)), f^3(\Phi^3(x_3))), f^{4,5}(f^4(\Phi^4(x_4)), f^5(\Phi^5(x_5))))$$

where $\Phi^\nu(x_\nu) = (\phi_{i_\nu}^\nu(x_\nu))_{i_\nu \in I^\nu} \in \mathbb{R}^{\#I^\nu}$.

Tree tensor networks as feed-forward neural networks

It corresponds to a **sum-product feed forward neural network** with a sparse architecture (given by T), a **number of hidden layers** equal to $\text{depth}(T) + 1$ (including a featuring layer), and **width** at level ℓ related to the α -ranks of the nodes α of level ℓ .

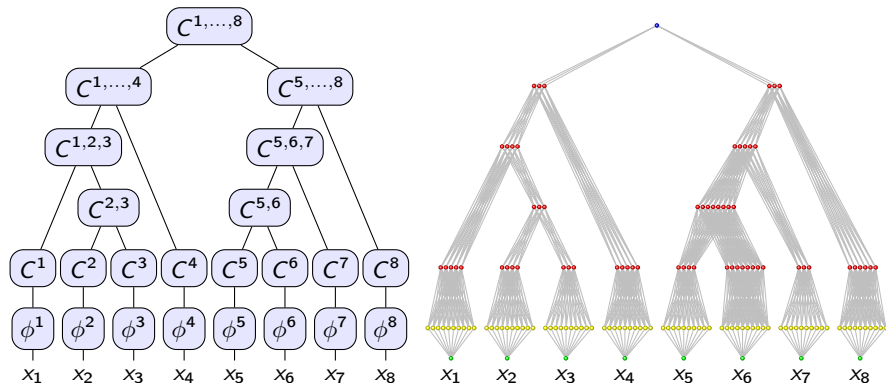


Figure: Tree tensor network and corresponding feed-forward sum-product neural network with 10 features per variable x_ν (right)

Properties of tree-based tensor formats

Many favorable properties inherited from the matrix case.

- Complexity is linear in d and polynomial in the rank for storage, evaluation, differentiation, integration...

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- Not so nonlinear approximation tool. A tensor u in tree-based format admits a multilinear parametrization with parameters $(C_\alpha)_{\alpha \in T}$ forming a tree tensor network, i.e.

$$u = R((C_\alpha)_{\alpha \in T})$$

with R a multilinear map.

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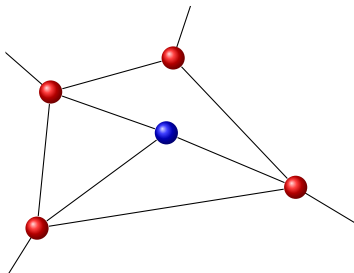
- **Topological properties** ensure the well-posedness of optimization problems and existence of **stable algorithms**
- **Geometrical properties** can be exploited for optimization and dynamical approximation.
- Possible extensions of **singular value decomposition** for u in a Hilbert tensor space V , and a way to obtain approximations u_r in $\mathcal{T}_r^T(V)$ such that

$$\|u - u_r\| \leq C_d \inf_{v \in \mathcal{T}_r^T(V)} \|u - v\|$$

with $C_d \sim \sqrt{d}$.

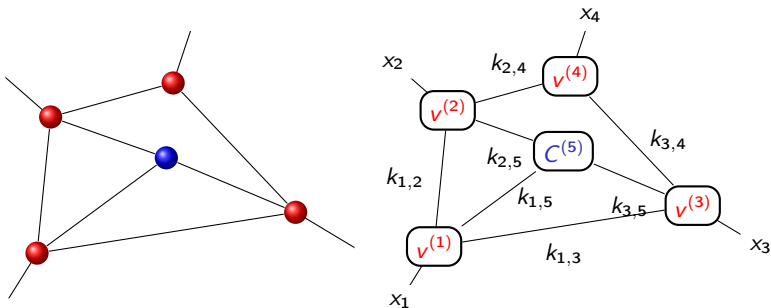
General tensor networks

More general tensor networks are associated with graphs $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ with nodes (vertices) $\mathcal{N}$ and edges $\mathcal{E}$, d of the nodes being associated with variables x_ν , $1 \leq \nu \leq d$



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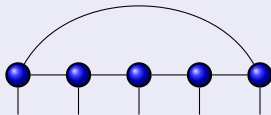
They have a **multilinear parametrization** of the form

$$v(x_1, \dots, x_d) = \sum_{\substack{1 \leq k_e \leq r_e \\ e \in \mathcal{E}}} \prod_{\nu=1}^d v^{(\nu)}(x_\nu, (k_e)_{e \in E_\nu}) \prod_{\nu=d+1}^N C^{(\nu)}((k_e)_{e \in E_\nu})$$

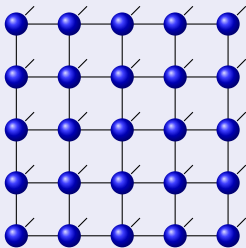
Tree tensor networks is a particular case where $\mathcal{G}$ is a tree.

Examples of tensor networks

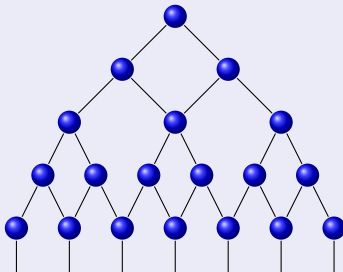
Tensor ring (MPS with periodic boundary conditions)



PEPS



MERA



When the graph contains cycles,

- integers r_e (bond dimensions) may not have an interpretation as α -ranks,
- no notion of singular value decomposition,
- loss of nice geometrical and topological properties,
- computational complexity increases,
- but yet powerful for some high-dimensional applications.

Outline

- 1 Tensors
- 2 Tensor ranks
- 3 Tensor networks
- 4 Tensorization**

Tensorization of vectors

A vector $\mathbf{v} \in \mathbb{R}^N$ with $N = b^L$ can be identified with a tensor of order L

$$\mathbf{v} \in \mathbb{R}^b \otimes \dots \otimes \mathbb{R}^b = (\mathbb{R}^b)^{\otimes L}$$

such that for $i \in \{0, \dots, N-1\}$

$$\mathbf{v}(i) = \mathbf{v}(i_1, \dots, i_L)$$

where $(i_1, \dots, i_L) \in \{0, \dots, b-1\}$ are the integers of the representation of i in base b

$$i = \sum_{k=1}^L i_k b^{L-k} = [i_1, \dots, i_L]_b.$$

The map which associates to $\mathbf{v}$ its tensorization $\mathbf{v}$ is a linear isometry from $\ell_2(\mathbb{R}^N)$ to $\ell_2(\mathbb{R}^b)^{\otimes L}$.

Some matrix-vector operations can be efficiently implemented using tensor algebra, such as the Hadamard transform

$$H_L \mathbf{v} \equiv (H_1 \otimes \dots \otimes H_1) \mathbf{v}$$

Tensorization of tensors

A tensor $\mathbf{v} \in \mathbb{R}^N \otimes \dots \otimes \mathbb{R}^N = (\mathbb{R}^N)^{\otimes d}$ with $N = b^L$ can be identified with a tensor of order dL

$$\mathbf{v} \in (\mathbb{R}^b)^{\otimes dL}$$

with

$$\mathbf{v}(\mathbf{i}_1, \dots, \mathbf{i}_d) = \mathbf{v}(i_1^1, \dots, i_1^L, \dots, i_d^1, \dots, i_d^L)$$

where

$$\mathbf{i}_\nu = [i_\nu^1 \dots i_\nu^{L_\nu}]_b$$

Other orderings of variables can be considered, such as

$$\mathbf{v}(\mathbf{i}_1, \dots, \mathbf{i}_d) = \mathbf{v}(i_1^1, \dots, i_d^1, \dots, i_1^L, \dots, i_d^L)$$

Tensors with different dimensions can be considered, i.e.

$$\mathbf{v} \in \mathbb{R}^{N_1} \otimes \dots \otimes \mathbb{R}^{N_d}, \quad N_\nu = b_\nu^{L_\nu}$$

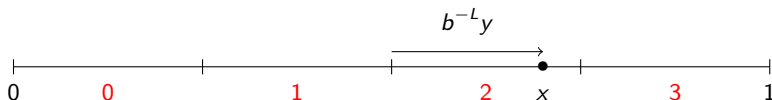
is identified with a tensor of order $\sum_{\nu=1}^d L_\nu$.

Tensorization of univariate functions

Consider a function $f \in \mathbb{R}^{[0,1)}$ defined on the interval $[0, 1)$.

- For $b, L \in \mathbb{N}$, we **subdivide uniformly** the interval $[0, 1)$ into b^L intervals. Any $x \in [0, 1)$ can be written

$$x = b^{-L}(i + y), \quad i \in \{0, \dots, b^L - 1\}, \quad y \in [0, 1).$$

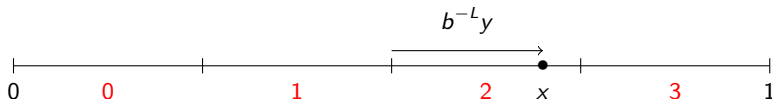


Tensorization of univariate functions

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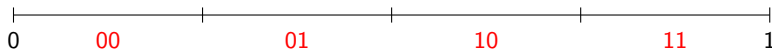
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- The integer i admits a **representation in base b**

$$i = \sum_{k=1}^L i_k b^{L-k} = [i_1 \dots i_L]_b, \quad i_k \in \{0, \dots, b-1\}$$

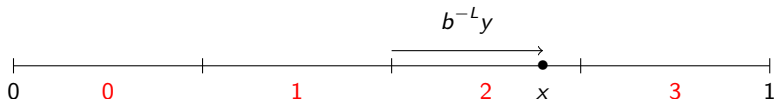


Tensorization of univariate functions

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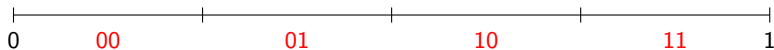
- For $b, L \in \mathbb{N}$, we **subdivide uniformly** the interval $[0, 1)$ into b^L intervals. Any $x \in [0, 1)$ can be written

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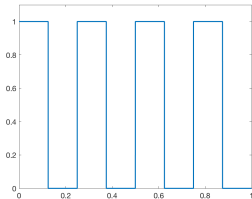


- f is thus identified with a **multivariate function (tensor of order $L + 1$)**

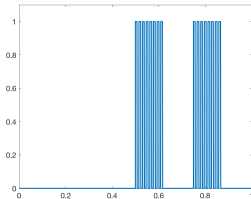
$$\mathbf{f} \in (\mathbb{R}^b)^{\otimes L} \otimes \mathbb{R}^{[0,1)} \quad \text{such that} \quad f(x) = \mathbf{f}(i_1, \dots, i_L, y)$$

Tensorization of univariate functions

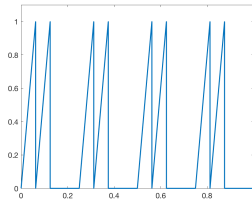
Examples of elementary tensors $f(x) = v^1(i_1) \dots v^L(i_L) v^{L+1}(y)$ ($b = 2$)



(a) $\delta_0(i_3)$



(b) $\delta_1(i_1)\delta_0(i_3)\delta_0(i_7)$



(c) $\delta_0(i_1)y$ ($L = 4$)

Polynomials

Consider a polynomial $q(x)$ of degree p . For any $\alpha \subset \{1, \dots, L\}$,

$$q(x) = q(b^{-L}(\sum_{k=1}^L i_k b^{L-k} + y)) = q(g(i_\alpha) + \tilde{g}(i_{\alpha^c})) = \sum_{j=0}^p g(i_\alpha)^j h_j(i_{\alpha^c})$$

so that $\text{rank}_\alpha(\mathbf{q}) \leq p + 1$.

Ranks of polynomials and splines

Polynomials

Consider a polynomial $q(x)$ of degree p . For any $\alpha \subset \{1, \dots, L\}$,

$$q(x) = q(b^{-L}(\sum_{k=1}^L i_k b^{L-k} + y)) = q(g(i_\alpha) + \tilde{g}(i_{\alpha^c})) = \sum_{j=0}^p g(i_\alpha)^j h_j(i_{\alpha^c})$$

so that $\text{rank}_\alpha(\mathbf{q}) \leq p + 1$.

Trigonometric polynomials

The tensorization of function $\cos(\omega x + \varphi)$ at resolution L has all ranks equal to 2.

Then a trigonometric polynomial $q(x)$ of degree p is such that for any $\alpha \subset \{1, \dots, L\}$,

$$\text{rank}_\alpha(\mathbf{q}) \leq 2p + 1.$$

Splines

A spline φ_N of degree p over N b -adic intervals forming a partition of $[0, 1)$ is such that

$$\text{rank}_{\{1, \dots, \nu\}}(\varphi_N) \leq \begin{cases} p + N, & 1 \leq \nu < \ell. \\ p + 1, & \ell \leq \nu \leq L. \end{cases}$$

where $b^{-\ell}$ is the minimal length of intervals.

Tensorization of multivariate functions

A function $f(x_1, \dots, x_d)$ defined on $[0, 1)^d$ can be similarly identified with a tensor of order $(L + 1)d$

$$\mathbf{f} \in (\mathbb{R}^b)^{\otimes Ld} \otimes (\mathbb{R}^{[0,1)})^{\otimes d}$$

such that

$$f(\mathbf{x}_1, \dots, \mathbf{x}_d) = \mathbf{f}(i_1^1, \dots, i_d^1, \dots, i_1^L, \dots, i_d^L, y_1, \dots, y_d)$$

$$\text{where } x_\nu = b^{-L} \left(\sum_{k=1}^L i_\nu^k b^{L-k} + y_\nu \right) = b^{-L} ([i_\nu^1 \dots i_\nu^L]_b + y_\nu)$$

The map $T_{b,d}$ which associates to a function f its tensorization $\mathbf{f}$ is a linear isometry from $L^p([0,1)^d)$ to $L^p(\{0,\dots,b-1\}^{L^d} \times [0,1)^d)$ for any $0 < p \leq \infty$.



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